

A Brief Introduction

This problem is set in an infinitely tall building, with our protagonists Alice and Bob on the ground floor and the N^{th} floor, respectively. Alice can board the lift and go up a_i floors, for every $a_i \in A$. The problem revolves around approximating the function $d_A(N)$, which measures the minimum number of steps Alice must take to reach Bob.

Conventions

We say $f \approx g$ if $\exists C \in \mathbb{R}$ such that

$$\frac{1}{C}g(N) - C \leq f(N) \leq Cg(N) + C$$

$\forall N \in \mathbb{N}$

Notice that $g \equiv 0 \implies -C \leq d_A(N) \leq C$, so any function approximately equal to 0 is bounded above, and vice versa.

Additionally, $A_t = \{\lfloor t^n \rfloor \mid n \geq 0\}$

Finally, let $A = \{1, a_2, a_3, \dots, a_n\}$ or $A = \{1, a_2, a_3, \dots\}$ with $1 < a_2 < a_3 < \dots$

Greedy Algorithm

The greedy algorithm is defined as the process of taking the largest possible step each time. If we determine that a set uses the greedy algorithm, we are done, since we have a formula for $d_A(N)$. Let

$$b_n = \left\lfloor \frac{N}{a_n} \right\rfloor, \text{ and}$$

$$b_i = \left\lfloor \frac{N - \sum_{k=i+1}^n a_k b_k}{a_i} \right\rfloor, \quad \forall i < n$$

Then, $d_A(N) = \sum_{k=0}^n b_k$. (Here, a_n is the largest $a_i \leq N$.)

Basic bounds on $d_A(N)$

- a) Notice that Alice can take 1 step at a time to go to any floor. Additionally, if Alice does so it takes N steps to reach N . So, $d_A(N) \leq N$.
- b) Assume for the sake of contradiction $d_A(M + N) > d_A(M) + d_A(N)$. Then, if we go M steps and then N steps we can reach $M + N$ in $d_A(M) + d_A(N)$ moves, which contradicts the minimality of d_A .

A is finite

Notice that

$$\frac{1}{a_n} \times \frac{N}{a_n} - a_n < \frac{N}{a_n} \leq d_A(N) \leq N < a_n \times \frac{N}{a_n} + a_n$$

so $d_A(N) \approx \frac{N}{a_n}$

$$A = \{1\} \cup \{P(n)\}_{n \in \mathbb{Z}_{\geq 0}}$$

Kamke's theorem states that if f is an integer valued polynomial with no fixed divisor ≥ 2 and for large enough s , the equation

$$f(x_1) + f(x_2) + \cdots + f(x_s) = N$$

has a solution for large enough N and in positive integers

$x_1, x_2, \dots, x_s$.

Now, let the largest fixed divisor of P be d . Let $\frac{P(n)}{d} = Q(n)$.

$P(x_1) + P(x_2) + \cdots + P(x_s) = d(Q(x_1) + Q(x_2) + \cdots + Q(x_s)) = dN$

has a solution for large enough N by Kamke's theorem, and now the remainder of the steps can be covered in $\leq d - 1$ moves.

$$a_i \geq h(i)$$

We claim that $h(i) = i!$ works. Notice that for $d_A(N) \geq \frac{N}{a_n}$ to be bounded, $\frac{a_{n+1}-1}{a_n} + 1 \geq \frac{a_{n+1}}{a_n}$ should also be bounded. (Here, a_n is the unique a_i such that $a_n \leq N < a_{n+1}$)

Now, assume for the sake of contradiction $\frac{a_2}{a_1} \leq C, \frac{a_3}{a_2} \leq C, \dots$

Multiplying until some n gives $\frac{n!}{a_n} \leq \frac{a_n}{a_1} \leq C^{n-1} \implies \left(\frac{n!}{a_1}\right)^{\frac{1}{n-1}}$ is unbounded.

It is well known that $n! \geq \left(\frac{n}{e}\right)^n$. Thus,

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{n!}{a_1}\right)^{\frac{1}{n-1}} &\geq \lim_{n \rightarrow \infty} \left(\frac{n^n}{e^n \cdot a_1}\right)^{\frac{1}{n-1}} \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{a_1}\right)^{\frac{1}{n-1}} \cdot \left(\frac{n}{e}\right)^{1+\frac{1}{n-1}} \\ &= \lim_{n \rightarrow \infty} \left(\frac{n}{e}\right) \lim_{n \rightarrow \infty} \left(\frac{n}{e \cdot a_1}\right)^{\frac{1}{n-1}} = \infty \end{aligned}$$

Is d_{A_p} ever unbounded?

We claim that there are infinitely many p that satisfy this property.

In fact, all $p : p \geq 2, p \in \mathbb{N}$ satisfy this property.

As proved in 4a), every natural number has a unique base p representation for $p \in \mathbb{N}$.

So, consider the number

$$p^k - 1 = (p - 1)p^{k-1} + (p - 1)p^{k-2} + \cdots + (p - 1).$$

$d_A(N) = k(p - 1) \rightarrow \infty$ as $k \rightarrow \infty$. So, d_{A_p} is unbounded as desired.

A_t for $t \in \mathbb{N}$

Notice that we can replace $(t + k)t^n$ with $t^{n+1} + k \cdot t^n$, so the coefficient of t^n is $\leq t - 1$. So, $d_{A_t}(N)$ is just the sum of the digits of N in base t .

$$d_{A_t} \not\approx d_{A_{t'}}$$

We claim that any subset of multiplicatively independent numbers work. To prove this, we assume for the sake of contradiction

$d_{A_p} \approx d_{A_q}$ for some multiplicatively independent p, q .

For the sake of contradiction assume that $\exists C$ such that

$$\frac{1}{C} d_{A_p}(N) - C \leq d_{A_q}(N) \leq C d_{A_p}(N) + C$$

Then,

$$\frac{1}{C} - \frac{C}{d_{A_p}(N)} \leq \frac{d_{A_q}(N)}{d_{A_p}(N)} \leq C + \frac{C}{d_{A_p}(N)}$$

$$d_{A_t} \not\approx d_{A_{t'}}$$

Let $N = p^n$

Thus, $d_{A_q}(p^n) \leq 2C \forall n$

By the Senge-Strauss Theorem, the number of integers the sum of whose digits in each of the bases p and q lies below a fixed bound is finite if and only if p and q are multiplicatively independent.

So, $d_{A_q}(p^n) + d_{A_p}(p^n) = d_{A_q}(p^n) + 1 \leq 2C + 1$ for finitely many n .
However, there are infinite choices for n , contradiction.

A as the set of Fibonacci numbers

Notice that we can make some assertions from the minimality of $d_A(N)$

- ▶ We can replace $f_{n-1} + f_n$ with f_{n+1} to decrease the number of moves.
- ▶ We can replace $2f_n$ with $f_{n+1} + f_{n-2}$, and $2f_1$ with 2.
- ▶ We can replace $3f_n$ with $f_{n-2} + f_{n+2}$ to strictly decrease the number of moves. (We can also replace $3f_1$ with 3)

So, we can reduce the problem to representing any number as the sum of non-consecutive Fibonacci numbers.

By Zeckendorf's theorem, we can represent every number uniquely as the sum of non consecutive Fibonacci numbers, using the greedy algorithm.

What if the building has basements?

Here, we tried to explore what would happen to d_A if we are also allowed to go down a_i floors $\forall a_i \in A$.

In this research direction, we deduced

- ▶ For d_{A_t} , $t \in \mathbb{N}$, we have that d_{A_t} is the sum of the digits of a number is balanced base t . (This is the unique representation of a number using digits from $\lceil -\frac{t}{2} \rceil$ to $\lfloor \frac{t}{2} \rfloor$)
- ▶ For a finite set, $d_A(N) \approx \frac{N}{a_n}$
- ▶ If $a_i \geq h(i)$, then d_A is still unbounded.

Generalization of Fibonacci numbers

Now, let us consider the set $\{1, h_2, h_3, \dots\}$. Here, $h_1 = 1, h_2 = k$ and $h_{n+1} = kh_n + h_{n-1}$

Notice that if we have kh_n and h_{n-1} , we can replace it with h_{n+1} .

So, if the coefficient of h_i is ever k , the coefficient of h_{i-1} is 0.

Additionally, we can have at most $(k-1)h_1$ since $kh_1 = h_2$.

V.E Hogatt Junior wrote a paper describing such sequences.

According to his work, every integer can be represented uniquely in this manner.

Unexplored research directions

- ▶ Given a function $f(n)$, when can you find a set A such that $d_A \approx f$.
- ▶ Given a set A , consider all subsets B such that $1 \in B$. Find or approximate the average of $d_B(N)$ over all such subsets B .